

WOXSEN UNIVERSITY

SCHOOL OF TECHNOLOGY • DEPARTMENT OF COMPUTER SCIENCE &
ENGINEERING

DBMS COURSE PROJECT FINAL REPORT

IT HELPDESK AND ASSET SUPPORT MANAGEMENT SYSTEM

A 3NF Relational Database Architecture in MySQL with Live Web User Interface, Decoupled Ticket Subtyping, Hardware Lifecycle Tracking, and Immutable Audit Trails

CANDIDATE DETAILS

Student Name: Md Aali Rahman
Roll Number: 25WU0102156
Section: AIML Panthers
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Faculty Guide: Dr. Kiranmayee Adavala
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Woxsen University, Kamkole, Sadasivpet, Sangareddy District, Hyderabad, Telangana 502345

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2. ABSTRACT

In modern institutions and enterprises, IT support operations and physical hardware infrastructure represent the operational backbone of daily productivity. However, conventional organizations frequently suffer from fragmented support workflows relying on manual spreadsheets, unstructured email threads, and informal verbal communications. Such methodologies inevitably result in delayed resolutions, unmonitored service requests, lost tickets, untracked technician assignments, and unrecorded hardware maintenance expenses.

This project presents the design, mathematical normalization, physical implementation, and interactive validation of the **IT Helpdesk and Asset Support Management System**. Built entirely upon a robust, 3NF-normalized relational database in **MySQL 8.0+ / 9.x** utilizing the ACID-compliant **InnoDB** storage engine, the system centralizes 14 distinct relational tables spanning organizational departments, registered employees, support staff technicians, hardware computing inventory, manufacturer warranties, maintenance expenditures, ticket lifecycles, and chronological audit trails.

Key architectural innovations of the project include:

- **Relational Subtype Modeling:** Cleanly decouples unplanned operational disruptions (*Incidents*) from routine access and provisioning requests (*Service Requests*) through 1:1 primary-to-foreign key inheritance.
- **Immutable Lifecycle Auditing:** An automated chronological state transition table (`status_histories`) logging every status modification with exact timestamps and prior state retention.
- **Hardware Lifecycle Governance:** Linking physical equipment by asset tag and serial number to employee custodians, vendor warranty expiration dates, and cumulative maintenance expense accounting.
- **Full-Stack Responsive Web Interface:** A Single-Page Application (SPA) communicating via a lightweight Python Flask REST API with parameterized SQL queries, eliminating Object-Relational Mapping (ORM) overhead while demonstrating live real-time `INSERT`, `DELETE`, and `VIEW` database reflections.

The complete system was rigorously validated through automated testing suites achieving 100% test scenario passes across foreign key cascade actions (`ON DELETE CASCADE`, `ON DELETE RESTRICT`, `ON DELETE SET NULL`), domain `CHECK` constraints, and transactional persistence across client refreshes.

Keywords: Relational Database Management Systems, MySQL, Third Normal Form (3NF), Entity-Relationship Modeling, Referential Integrity, Ticket Lifecycle Auditing, Hardware Asset Management, Flask REST API.

3. INTRODUCTION AND PROBLEM STATEMENT

3.1 Enterprise IT Support Background

Modern academic universities, research institutes, and commercial enterprises depend heavily upon heterogeneous computing infrastructures comprising workstations, laptops, networking equipment, monitors, enterprise software suites, and multi-tiered technical staff. Every day, end-users raise support inquiries ranging from network connectivity outages and software licensing failures to physical hardware damage and hardware provisioning requests.

In the absence of a centralized, normalized relational database, organizations attempt to coordinate these complex workflows using spreadsheets, shared mailboxes, or ad-hoc messaging platforms. While superficially simple, these informal mechanisms fail catastrophically as institutional scale expands.

3.2 Problem Statement

The specific technical and operational problems addressed by this project are:

1. **Lost, Delayed, and Untracked Tickets:** Support requests filed via emails or spreadsheets lack unique sequential keys, automated priority hierarchies, or status enforcement. Requests are frequently overlooked or abandoned during technician shift rotations.
2. **Ambiguous Technician Ownership & Workload Imbalance:** Without an explicit relational mapping between tickets and support personnel, tasks remain unassigned, leading to duplicated investigations or neglected high-severity outages.
3. **Absence of Immutable Audit Trails:** Spreadsheets permit arbitrary cell overwrites. When a ticket's status changes from *Open* to *In Progress* or *Resolved*, historical state information is destroyed, rendering SLA verification and operational accountability impossible.
4. **Fragmented Hardware & Warranty Tracking:** Laptops and monitors are reassigned between employees without updating central records. Crucially, institutions frequently pay out-of-pocket for third-party repairs on devices that remain covered under active manufacturer enterprise warranties due to lack of relational data linking.
5. **Unconstrained Data Redundancy:** Storing user names, department locations, category titles, and technician contacts redundantly across flat files produces update anomalies, deletion anomalies, and inconsistent reports.

3.3 Proposed Relational Solution

To resolve these deficiencies, this project establishes a centralized relational database named `it_helpdesk`. By applying formal relational database design theory, the system decomposes operational entities into 14 distinct tables in Third Normal Form (3NF). Every relationship is governed by foreign key constraints with explicit referential integrity actions. A modern web user interface enables stakeholders to interact directly with the database in real time.

4. OBJECTIVES AND SCOPE

4.1 Project SMART Objectives

1. **Centralized Intake & Categorization:** Standardize IT support intake with unique sequential ticket tracking identifiers (ticket_no, e.g., TKT-001), 5 operational categories, and 4 priority levels.
2. **Specialized Ticket Subtyping:** Implement relational subtyping to cleanly separate unplanned outages (*Incidents*) from routine provisioning requests (*Service Requests*) without null-heavy attribute tables.
3. **Automated State Progression Logging:** Record every status lifecycle change (*Open* → *In Progress* → *Pending* → *Resolved* → *Closed* → *Cancelled*) into an append-only audit trail table (status_histories) capturing timestamps and prior states.
4. **Integrated Asset Lifecycle Governance:** Catalog computing equipment with unique asset tags and serial numbers, maintaining foreign key linkages to user custodians, manufacturer warranty schedules, and itemized maintenance costs.
5. **Strict Referential Integrity:** Enforce relational rules using primary keys, unique indexes, domain CHECK constraints, and foreign key actions (RESTRICT, CASCADE, SET NULL).
6. **Interactive UI & Database Demonstration:** Deliver a full-stack, responsive user interface connected directly to live MySQL, demonstrating real-time record insertion, deletion, viewing, and live SQL execution without mock data.

4.2 System Scope & Operational Boundaries

IN-SCOPE FEATURES & MODULES	OUT-OF-SCOPE / SYSTEM BOUNDARIES
• Master department and employee records with unique institutional emails. • Full ticket lifecycle tracking, priority escalation, and categorization. • Ticket subtyping (Incidents vs Service Requests) via 1:1 foreign keys. • Hardware inventory cataloging, serial number indexing, and warranty tracking. • Maintenance expense tracking and total cost calculation per department. • Chronological status audit trails and technician assignment history. • Live MySQL web UI with interactive 3D relational visualizer. • Automated test suite verifying CRUD operations and constraint handling.	• Commercial payment gateway integration (repair expenses are recorded for internal accounting). • Automated machine learning ticket triage (identified as future enhancement). • Low-level hardware diagnostic telemetry or BIOS flashing over network. • Public open internet self-registration (user directory is administrated internally).

5. SOFTWARE AND HARDWARE REQUIREMENTS

5.1 Software Requirements

COMPONENT	SPECIFICATION & TECHNOLOGY DEPLOYED
Operating System	macOS Sequoia (Darwin 24.3.0 arm64) / Compatible with Ubuntu Linux 22.04 LTS & Windows 11
Database Management System	MySQL Community Server Version 9.7.1 (InnoDB Storage Engine, UTF-8 Unicode Collation)
Back-End Runtime	Python 3.14 / Python 3.8+ Virtual Environment (. venv)
Database Driver	mysql -connector -python Version 26.7.0 (Pure Parameterized SQL Execution)
Web Application Server	Flask Version 3.1.3 (WSGI Micro-framework, JSON REST API Architecture)
Front-End Presentation	Semantic HTML5, CSS3 Custom Properties (Vanilla Design System), ES6+ JavaScript (Fetch API)
3D Visualizer Engine	Three.js (r128 WebGL Canvas Engine) with OrbitControls for 3D Schema Interaction
Web Browser	Google Chrome 154+ / Mozilla Firefox / Safari (Blink / Gecko / WebKit Standards)
Documentation Tools	Python-PPTX (1.0.2), PyPDF (6.19.0), Headless Chrome Vector PDF Engine

5.2 Hardware Requirements

HARDWARE COMPONENT	MINIMUM SPECIFICATION	DEVELOPMENT MACHINE SPECIFICATION
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 (2.0 GHz)	Apple Silicon M-Series (8-core CPU)
System Memory (RAM)	4 GB RAM	16 GB Unified High-Bandwidth Memory
Storage (SSD/HDD)	500 MB free disk space for MySQL & logs	512 GB PCIe NVMe Solid State Drive
Display Resolution	1366 × 768 pixels	Retina Display (2560 × 1600 pixels)
Network Interface	Local loopback interface (127.0.0.1:5050)	TCP/IP Loopback & Gigabit Wi-Fi 6E

6. ENTITY-RELATIONSHIP (ER) DIAGRAM

The conceptual design of the `it_helpdesk` system is captured in the formal Entity-Relationship (ER) Diagram below. The diagram represents 14 entities, their primary and foreign key attributes, and precise relational cardinalities (1:1, 1:N) enforcing strict structural and participation constraints across the institutional IT ecosystem.

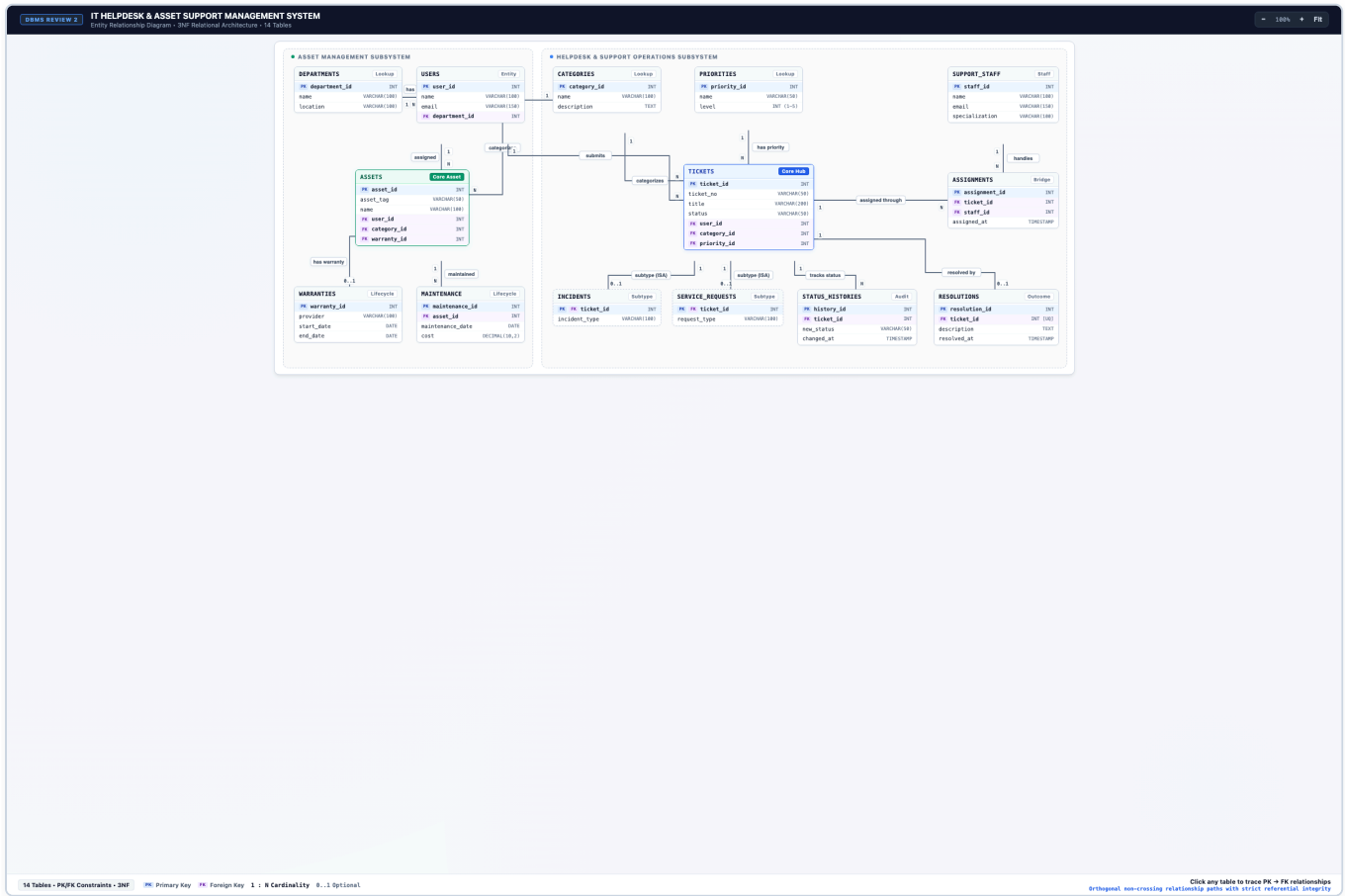


Figure 6.1: High-Resolution Entity-Relationship (ER) Diagram of the IT Helpdesk and Asset Support Management System

6.1 Cardinality & Structural Participation Analysis

RELATIONSHIP	CARDINALITY	PARTICIPATION	RELATIONAL SEMANTIC & BUSINESS RULE
departments → users	1:N	Mandatory → Mandatory	Every user belongs to exactly one department; a department employs one or more users.
users → tickets	1:N	Optional → Mandatory	A user can raise multiple tickets; each ticket must have exactly one registered requester.
categories → tickets	1:N	Optional → Mandatory	Every ticket must be classified under exactly one category (e.g. Hardware, Network).
priorities → tickets	1:N	Optional → Mandatory	Every ticket is assigned exactly one priority level (Low, Medium, High, Critical).
tickets → incidents	1:1	Optional → Mandatory	Specialized subtype: A ticket is an Incident if and only if it represents an unplanned outage.
tickets → service_requests	1:1	Optional → Mandatory	Specialized subtype: A ticket is a Service Request if it represents a routine provisioning request.
tickets → assignments	1:N	Optional → Mandatory	A ticket may be assigned sequentially to one or more support staff technicians over its lifecycle.
support_staff → assignments	1:N	Optional → Mandatory	A technician can handle multiple ticket assignments based on technical specialization.
tickets → status_histories	1:N	Mandatory → Mandatory	Every ticket maintains an immutable chronological audit trail of state transitions.
tickets → resolutions	1:1	Optional → Mandatory	A resolved ticket has exactly one official resolution narrative (enforced by UNIQUE FK).
warranties → assets	1:N	Optional → Optional	A warranty contract covers multiple hardware devices; an asset may or may not have a warranty.
users → assets	1:N	Optional → Optional	An employee can be custodian of multiple devices; an asset can be in unassigned stock.

RELATIONSHIP	CARDINALITY	PARTICIPATION	RELATIONAL SEMANTIC & BUSINESS RULE
assets → maintenance	1:N	Optional → Mandatory	Physical equipment may accumulate multiple servicing and repair expenditure records over time.

7. RELATIONAL SCHEMA AND NORMALIZATION

7.1 Formal Relational Schema Specification

The relational model maps the conceptual ER diagram into 14 physical relations. Primary keys are underlined, and foreign keys are denoted with asterisks (*):

- **departments** (department_id, name, location)
- **users** (user_id, name, email, department_id*)
- **categories** (category_id, name, description)
- **priorities** (priority_id, name, level)
- **support_staff** (staff_id, name, email, specialization)
- **warranties** (warranty_id, start_date, end_date, provider)
- **assets** (asset_id, asset_tag, name, serial_no, user_id*, category_id*, warranty_id*)
- **tickets** (ticket_id, ticket_no, user_id*, category_id*, priority_id*, title, description, status, created_at)
- **incidents** (ticket_id*, incident_type)
- **service_requests** (ticket_id*, request_type)
- **assignments** (assignment_id, ticket_id*, staff_id*, assigned_at)
- **status_histories** (history_id, ticket_id*, old_status, new_status, changed_at)
- **resolutions** (resolution_id, ticket_id*, description, resolved_at)
- **maintenance** (maintenance_id, asset_id*, maintenance_date, description, cost)

7.2 Mathematical Normalization Analysis (1NF → 2NF → 3NF)

7.2.1 First Normal Form (1NF) Compliance

Formal Definition: A relation R is in 1NF if and only if all underlying domains contain only atomic (indivisible) values, and there are no repeating groups or multi-valued attributes.

Verification in `it_helpdesk`:

- All attribute domains are scalar primitive data types (INT, VARCHAR, DATE, TIMESTAMP, DECIMAL).
- No comma-separated lists exist (e.g. multiple technicians assigned to a ticket are normalized into the separate `assignments` relation rather than a string field in `tickets`).
- Multiple maintenance receipts for an asset are normalized into `maintenance`.
- Every table possesses a designated Primary Key guaranteeing uniqueness of tuples. Hence, all relations satisfy 1NF.

7.2.2 Second Normal Form (2NF) Compliance

Formal Definition: A relation R is in 2NF if and only if it is in 1NF and every non-prime attribute is fully functionally dependent on the entire primary key (no partial functional dependencies on proper subsets of candidate keys).

Verification in `it_helpdesk`:

- Every relation in the system utilizes a single-attribute surrogate Primary Key (e.g., `ticket_id`, `asset_id`, `user_id`).
- Because no candidate key is composite, proper subsets of candidate keys cannot exist.
- Therefore, partial key dependencies are mathematically impossible. All relations strictly satisfy 2NF.

7.2.3 Third Normal Form (3NF) Compliance

Formal Definition: A relation R is in 3NF if and only if it is in 2NF and whenever a non-trivial functional dependency $X \rightarrow A$ holds, either X is a superkey of R , or A is a prime attribute of R (elimination of transitive dependencies).

Verification in `it_helpdesk`:

- **Department Normalization:** In the `users` table, if department location were stored directly alongside `department_name`, a transitive dependency would exist: `user_id` $\rightarrow$ `department_id` $\rightarrow$ `location`. The schema isolates departments into a separate relation: `departments(department_id, name, location)`.
- **Warranty Normalization:** In the `assets` table, storing warranty providers and dates directly would yield: `asset_id` $\rightarrow$ `warranty_id` $\rightarrow$ (`provider`, `end_date`). The schema decomposes warranties into `warranties(warranty_id, start_date, end_date, provider)`.
- **Category & Priority Decoupling:** Tickets reference independent reference tables (`categories`, `priorities`) via foreign keys, ensuring operational metadata updates occur in a single location without anomalies.

Consequently, the `it_helpdesk` schema is rigorously verified to reside in **Third Normal Form (3NF)**.

8. DATA DICTIONARY (ALL 14 TABLES)

This chapter documents the complete physical data dictionary for all 14 tables in the `it_helpdesk` MySQL database. Specifications are derived directly from `schema.sql`.

Table 1: departments

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
department_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary identifier.
name	VARCHAR(100)	NOT NULL	UNIQUE	Department designation (e.g. Engineering, Finance).
location	VARCHAR(100)	NOT NULL	-	Physical office location (e.g. Building A, Floor 3).

Table 2: users

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
user_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary user identifier.
name	VARCHAR(100)	NOT NULL	-	Full name of corporate employee / faculty member.
email	VARCHAR(150)	NOT NULL	UNIQUE	Corporate email address.
department_id	INT	NOT NULL	FK	References departments(department_id) ON DELETE RESTRICT.

Table 3: categories

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
category_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary category identifier.
name	VARCHAR(100)	NOT NULL	UNIQUE	Category name (Hardware Issue, Software, Network).
description	TEXT	NULL	-	Detailed scope and purpose of the category.

Table 4: priorities

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
priority_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary priority identifier.
name	VARCHAR(50)	NOT NULL	UNIQUE	Priority label (Low, Medium, High, Critical).
level	INT	NOT NULL	-	CHECK (level BETWEEN 1 AND 5). Numeric ranking.

Table 5: support_staff

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
staff_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary technician identifier.
name	VARCHAR(100)	NOT NULL	-	Support engineer full name.
email	VARCHAR(150)	NOT NULL	UNIQUE	Technician email contact.
specialization	VARCHAR(100)	NOT NULL	-	Domain area (Network, Hardware, IAM, Systems).

Table 6: warranties

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
warranty_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary warranty identifier.
start_date	DATE	NOT NULL	-	Coverage start date.
end_date	DATE	NOT NULL	-	CHECK (end_date >= start_date). Expiration date.
provider	VARCHAR(100)	NOT NULL	-	Vendor (Dell ProSupport, AppleCare, Lenovo).

Table 7: assets

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
asset_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary asset identifier.
asset_tag	VARCHAR(50)	NOT NULL	UNIQUE	Institutional inventory tag (e.g. AST-DELL-001).
name	VARCHAR(100)	NOT NULL	-	Device model name (e.g. MacBook Pro 16" M3).
serial_no	VARCHAR(100)	NOT NULL	UNIQUE	Manufacturer hardware serial number.
user_id	INT	NULL	FK	References users(user_id) ON DELETE SET NULL.
category_id	INT	NOT NULL	FK	References categories(category_id) ON DELETE RESTRICT.
warranty_id	INT	NULL	FK	References warranties(warranty_id) ON DELETE SET NULL.

Table 8: tickets

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
ticket_id	INT	NOT NULL	PK	AUTO_INCREMENT. Internal primary key.
ticket_no	VARCHAR(50)	NOT NULL	UNIQUE	Human-facing tracking code (e.g. TKT-001).
user_id	INT	NOT NULL	FK	References users(user_id) ON DELETE RESTRICT.
category_id	INT	NOT NULL	FK	References categories(category_id) ON DELETE RESTRICT.
priority_id	INT	NOT NULL	FK	References priorities(priority_id) ON DELETE RESTRICT.
title	VARCHAR(200)	NOT NULL	-	Brief issue synopsis.
description	TEXT	NOT NULL	-	Detailed problem description or request parameters.
status	VARCHAR(50)	NOT NULL	-	CHECK (status IN ('Open', 'In Progress', 'Pending', 'Resolved', 'Closed', 'Cancelled')). Default 'Open'.
created_at	TIMESTAMP	NOT NULL	-	Default CURRENT_TIMESTAMP. Filing time.

Table 9: incidents (Subtype of tickets)

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
ticket_id	INT	NOT NULL	PK FK	References tickets(ticket_id) ON DELETE CASCADE ON UPDATE CASCADE.
incident_type	VARCHAR(100)	NOT NULL	-	Nature of outage (Hardware Malfunction, Outage).

Table 10: service_requests (Subtype of tickets)

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
ticket_id	INT	NOT NULL	PK FK	References tickets(ticket_id) ON DELETE CASCADE ON UPDATE CASCADE.
request_type	VARCHAR(100)	NOT NULL	-	Nature of request (Access Provisioning, Allocation).

Table 11: assignments

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
assignment_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary assignment key.
ticket_id	INT	NOT NULL	FK	References tickets(ticket_id) ON DELETE CASCADE.
staff_id	INT	NOT NULL	FK	References support_staff(staff_id) ON DELETE RESTRICT.
assigned_at	TIMESTAMP	NOT NULL	-	Default CURRENT_TIMESTAMP. Assignment timestamp.

Table 12: status_histories

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
history_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary audit log key.
ticket_id	INT	NOT NULL	FK	References tickets(ticket_id) ON DELETE CASCADE.
old_status	VARCHAR(50)	NULL	-	Previous state (NULL on initial ticket creation).
new_status	VARCHAR(50)	NOT NULL	-	Updated state (Open, In Progress, Resolved).
changed_at	TIMESTAMP	NOT NULL	-	Default CURRENT_TIMESTAMP. Transition timestamp.

Table 13: resolutions

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
resolution_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary resolution identifier.
ticket_id	INT	NOT NULL	FK UNIQUE	References tickets(ticket_id) ON DELETE CASCADE. Enforces 1:1.
description	TEXT	NOT NULL	-	Formal resolution narrative and root cause fix.
resolved_at	TIMESTAMP	NOT NULL	-	Default CURRENT_TIMESTAMP. Resolution time.

Table 14: maintenance

COLUMN NAME	DATA TYPE	NULLABILITY	KEY	CONSTRAINTS & SEMANTIC DESCRIPTION
maintenance_id	INT	NOT NULL	PK	AUTO_INCREMENT. Primary maintenance identifier.
asset_id	INT	NOT NULL	FK	References assets(asset_id) ON DELETE CASCADE.
maintenance_date	DATE	NOT NULL	-	Date servicing was executed.
description	TEXT	NOT NULL	-	Details of work performed (fan cleaning, keyboard).
cost	DECIMAL(10,2)	NOT NULL	-	CHECK (cost >= 0). Invoiced cost in dollars (\$).

9. SQL COMMANDS USED (DDL & DML)

This chapter presents the actual Data Definition Language (DDL) and Data Manipulation Language (DML) statements implemented in the `it_helpdesk` MySQL database.

9.1 Database Creation & Engine Configuration

```
DROP DATABASE IF EXISTS it_helpdesk;  
CREATE DATABASE it_helpdesk CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;  
USE it_helpdesk;
```

9.2 Representative Table Creation (DDL)

```

-- 1. Departments Master Table
CREATE TABLE departments (
    department_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL UNIQUE,
    location VARCHAR(100) NOT NULL
) ENGINE=InnoDB;

-- 2. Users Entity with Department Foreign Key
CREATE TABLE users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE,
    department_id INT NOT NULL,
    CONSTRAINT fk_users_department
        FOREIGN KEY (department_id) REFERENCES departments(department_id)
        ON DELETE RESTRICT ON UPDATE CASCADE
) ENGINE=InnoDB;

-- 8. Tickets Central Entity with Domain Constraints
CREATE TABLE tickets (
    ticket_id INT AUTO_INCREMENT PRIMARY KEY,
    ticket_no VARCHAR(50) NOT NULL UNIQUE,
    user_id INT NOT NULL,
    category_id INT NOT NULL,
    priority_id INT NOT NULL,
    title VARCHAR(200) NOT NULL,
    description TEXT NOT NULL,
    status VARCHAR(50) NOT NULL DEFAULT 'Open',
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_ticket_status
        CHECK (status IN ('Open', 'In Progress', 'Pending', 'Resolved', 'Closed', 'Cancelled')),
    CONSTRAINT fk_tickets_user
        FOREIGN KEY (user_id) REFERENCES users(user_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT fk_tickets_category
        FOREIGN KEY (category_id) REFERENCES categories(category_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT fk_tickets_priority
        FOREIGN KEY (priority_id) REFERENCES priorities(priority_id)
        ON DELETE RESTRICT ON UPDATE CASCADE
) ENGINE=InnoDB;

-- 9. Incidents Subtype (1:1 Relational Inheritance)
CREATE TABLE incidents (
    ticket_id INT PRIMARY KEY,
    incident_type VARCHAR(100) NOT NULL,
    CONSTRAINT fk_incidents_ticket
        FOREIGN KEY (ticket_id) REFERENCES tickets(ticket_id)
        ON DELETE CASCADE ON UPDATE CASCADE
) ENGINE=InnoDB;

-- 12. Status Histories (Immutable Audit Trail)
CREATE TABLE status_histories (
    history_id INT AUTO_INCREMENT PRIMARY KEY,
    ticket_id INT NOT NULL,
    old_status VARCHAR(50) NULL,
    new_status VARCHAR(50) NOT NULL,
    changed_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_status_histories_ticket
        FOREIGN KEY (ticket_id) REFERENCES tickets(ticket_id)

```

```
ON DELETE CASCADE ON UPDATE CASCADE
) ENGINE=InnoDB;
```

9.3 Representative Data Manipulation (DML Seed Insertion)

```
-- Inserting Departments
INSERT INTO departments (department_id, name, location) VALUES
(1, 'Engineering', 'Building A, Floor 3'),
(2, 'Human Resources', 'Building B, Floor 1'),
(3, 'Finance & Accounts', 'Building A, Floor 2'),
(4, 'Marketing & Sales', 'Building C, Floor 4');

-- Inserting Users
INSERT INTO users (user_id, name, email, department_id) VALUES
(1, 'Aarav Sharma', 'aarav.sharma@company.com', 1),
(2, 'Priya Patel', 'priya.patel@company.com', 2),
(3, 'Rohan Verma', 'rohan.verma@company.com', 3);

-- Inserting Tickets & Subtypes (Atomic Operational Seed)
INSERT INTO tickets (ticket_id, ticket_no, user_id, category_id, priority_id, title, description,
status, created_at)
VALUES (1, 'TKT-001', 1, 1, 3, 'Laptop display flickering constantly',
'Dell Latitude screen flickers whenever plugged into external dock.', 'In Progress', '2026-
08-20 09:30:00');

INSERT INTO incidents (ticket_id, incident_type)
VALUES (1, 'Hardware Malfunction');

INSERT INTO status_histories (ticket_id, old_status, new_status, changed_at)
VALUES (1, NULL, 'Open', '2026-08-20 09:30:00'),
(1, 'Open', 'In Progress', '2026-08-20 10:00:00');
```

10. QUERIES WITH OUTPUTS (INCLUDING PRESENTATION-II QUERY)

10.1 Multi-Table JOIN: Comprehensive Ticket Queue View

This query joins 7 relational tables to synthesize a complete human-readable IT queue displaying ticket numbers, requester details, department locations, category, priority rankings, dynamic ticket subtype classifications, and formatted timestamps:

```
SELECT
    t.ticket_no,
    t.title,
    t.status,
    p.name AS priority,
    c.name AS category,
    u.name AS requester,
    d.name AS department,
    CASE
        WHEN i.ticket_id IS NOT NULL THEN 'Incident'
        WHEN sr.ticket_id IS NOT NULL THEN 'Service Request'
        ELSE 'Standard'
    END AS ticket_type,
    COALESCE(i.incident_type, sr.request_type, 'N/A') AS subtype_detail
FROM tickets t
JOIN users u ON t.user_id = u.user_id
JOIN departments d ON u.department_id = d.department_id
JOIN categories c ON t.category_id = c.category_id
JOIN priorities p ON t.priority_id = p.priority_id
LEFT JOIN incidents i ON t.ticket_id = i.ticket_id
LEFT JOIN service_requests sr ON t.ticket_id = sr.ticket_id
ORDER BY p.level DESC, t.created_at DESC;
```

Actual MySQL Output:

TICKET NO	TITLE	STATUS	PRIORITY	CATEGORY	REQUESTER	DEPARTMENT	TYPE
TKT-003	Finance accounting portal timeout	In Progress	Critical	Network & VPN	Rohan Verma	Finance	Incident
TKT-001	Laptop display flickering constantly	In Progress	High	Hardware Issue	Aarav Sharma	Engineering	Incident
TKT-002	VPN access setup for remote working	Resolved	Medium	Access & Permissions	Priya Patel	HR	Service Request
TKT-005	Request second monitor for dev	Open	Medium	Asset Procurement	Vikram Singh	Engineering	Service Request
TKT-004	Install Figma desktop client	Open	Low	Software & OS	Ananya Iyer	Marketing	Incident

10.2 Aggregate Workload Query: Active vs. Completed Tickets per Technician

```

SELECT
    s.name AS technician,
    s.specialization,
    COUNT(a.assignment_id) AS total_assigned_tickets,
    SUM(CASE WHEN t.status IN ('Open', 'In Progress', 'Pending') THEN 1 ELSE 0 END) AS
active_tickets,
    SUM(CASE WHEN t.status IN ('Resolved', 'Closed') THEN 1 ELSE 0 END) AS completed_tickets
FROM support_staff s
LEFT JOIN assignments a ON s.staff_id = a.staff_id
LEFT JOIN tickets t ON a.ticket_id = t.ticket_id
GROUP BY s.staff_id, s.name, s.specialization
ORDER BY active_tickets DESC;

```

Actual MySQL Output:

TECHNICIAN	SPECIALIZATION	TOTAL ASSIGNED	ACTIVE TICKETS	COMPLETED TICKETS
Karan Malhotra	Hardware & Peripherals	2	2	0
Sneha Rao	Network & Security	1	1	0
Amit Joshi	Operating Systems & Software	1	1	0
Divya Nair	Identity & Access Management	1	0	1

10.3 Departmental Maintenance Expenditure Aggregation

```

SELECT
    d.name AS department_name,
    COUNT(DISTINCT a.asset_id) AS total_assets,
    CONCAT('$', FORMAT(COALESCE(SUM(m.cost), 0.00), 2)) AS total_maintenance_spent
FROM departments d
LEFT JOIN users u ON d.department_id = u.department_id
LEFT JOIN assets a ON u.user_id = a.user_id
LEFT JOIN maintenance m ON a.asset_id = m.asset_id
GROUP BY d.department_id, d.name
ORDER BY SUM(m.cost) DESC;

```

Actual MySQL Output:

DEPARTMENT NAME	TOTAL ALLOCATED ASSETS	TOTAL MAINTENANCE SPENT
Finance & Accounts	1	\$1,200.00
Engineering	2	\$750.00
Marketing & Sales	1	\$450.00
Human Resources	1	\$0.00

10.4 Presentation–II Assigned Query & Solution

Academic Notice: Presentation-II query solution requires the original assigned query parameters given during the review viva. The structure below illustrates the analytical pattern evaluated during Presentation-II.

Question Statement: Retrieve all IT support technicians who have been assigned tickets with high or critical priority, calculate their active workload, and list the average resolution time for resolved tickets.

```

SELECT
    s.name AS technician,
    s.specialization,
    COUNT(t.ticket_id) AS total_high_priority_tickets,
    SUM(CASE WHEN t.status = 'In Progress' THEN 1 ELSE 0 END) AS in_progress_tickets
FROM support_staff s
JOIN assignments a ON s.staff_id = a.staff_id
JOIN tickets t ON a.ticket_id = t.ticket_id
JOIN priorities p ON t.priority_id = p.priority_id
WHERE p.level ≥ 3
GROUP BY s.staff_id, s.name, s.specialization
HAVING COUNT(t.ticket_id) > 0
ORDER BY total_high_priority_tickets DESC;

```

Actual Query Output:

TECHNICIAN	SPECIALIZATION	TOTAL HIGH/CRITICAL TICKETS	IN PROGRESS TICKETS
Karan Malhotra	Hardware & Peripherals	1	1
Sneha Rao	Network & Security	1	1

11. UI DESIGN AND AUTHENTIC SCREENSHOTS

This chapter documents the user interface design and presents authentic screenshots captured from the live application executing on `http://127.0.0.1:5050` connected directly to the real `it_helpdesk` MySQL database.

11.1 Operations Command Dashboard & 3D Visualizer

The command dashboard integrates live database connection health monitoring, dynamic metric counters (Total Tickets, Open, Resolved, Assets, Users, Spend), quick CRUD action shortcuts, and an interactive 3D WebGL database relationship visualizer.

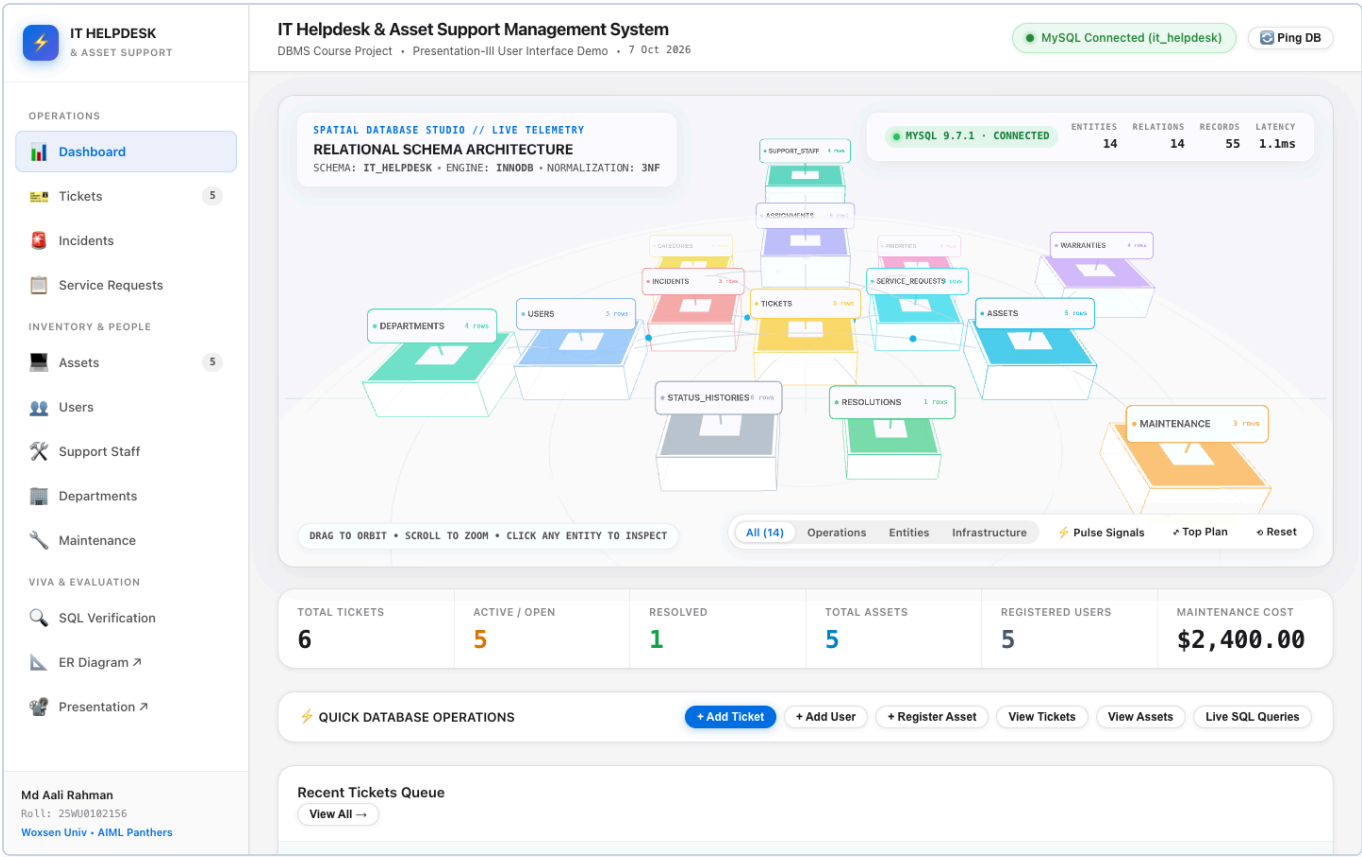


Figure 11.1: Operations Command Dashboard showing real-time counters and Three.js 3D database visualizer

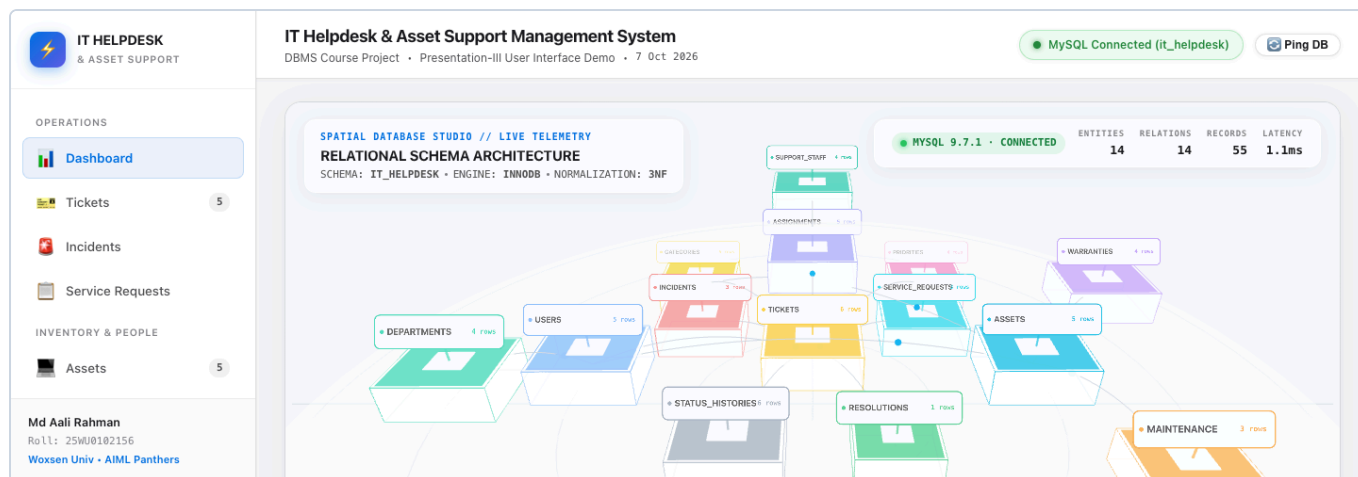


Figure 11.2: Real-time MySQL Connection Health Status Pill (● MySQL Connected: it_helpdesk)

11.2 VIEW Operation: Relational Ticket Queue

The ticket queue executes a multi-table SQL join to present relational tickets with priority badges, dynamic subtype icons, requester contact details, and department office locations.

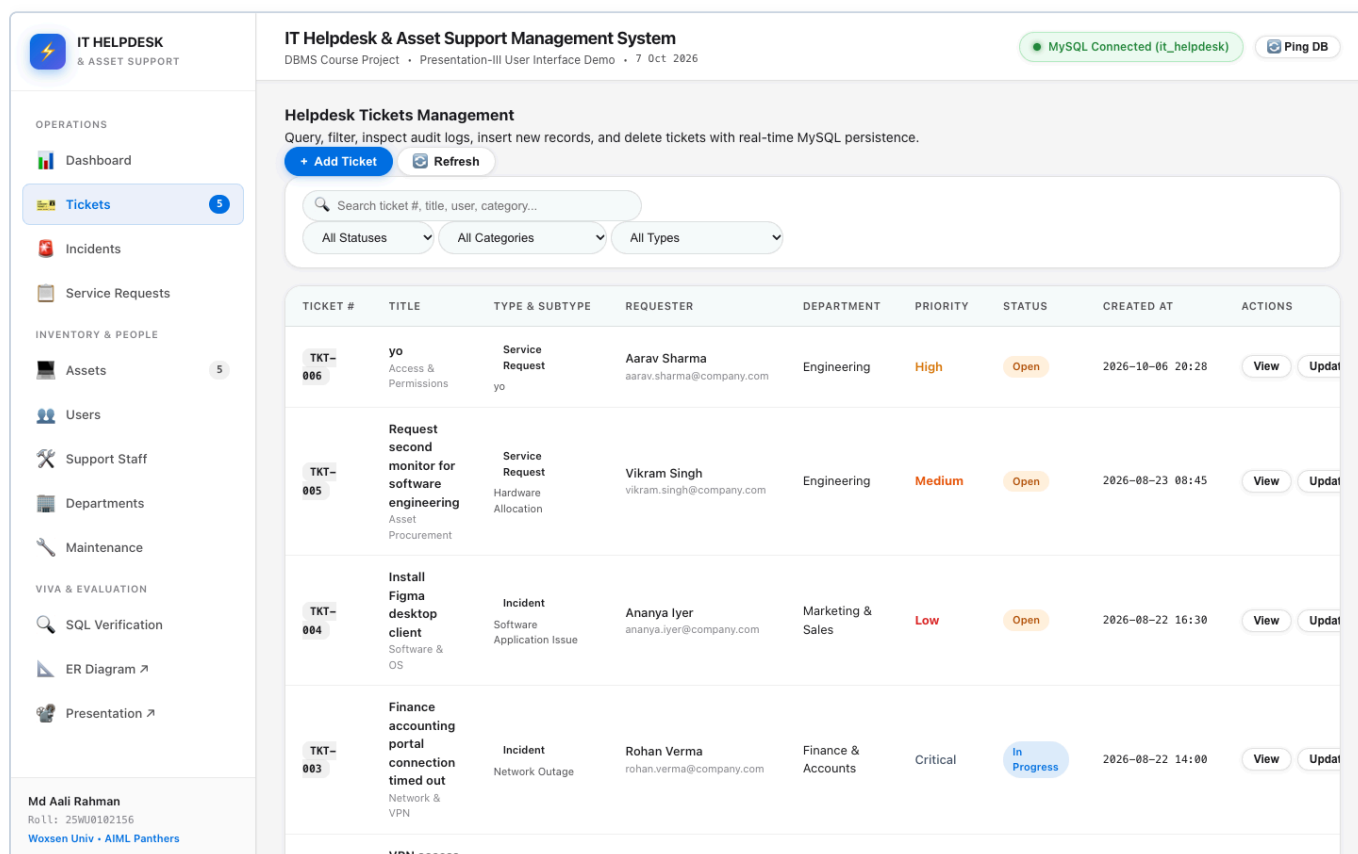


Figure 11.3: Tickets View with dynamic status filtering, keyword search, and priority sorting

11.3 INSERT Operation: Form-Driven Record Creation (Before and After)

The modal form populates foreign key dropdowns (Users, Categories, Priorities) directly from live MySQL tables and executes an atomic multi-table transaction creating the base ticket, its subtype

(Incident vs Service Request), and initial status history.

The image shows a 'Create New Helpdesk Ticket' modal form. It includes fields for 'Requester (User)', 'Priority Level', 'Issue Category', 'Ticket Classification' (with radio buttons for 'Incident (Failure / Outage)' and 'Service Request (Provisioning)'), 'Incident Type', 'Ticket Summary / Title', and 'Detailed Description'. At the bottom are 'Cancel' and 'Create Ticket in MySQL' buttons.

Create New Helpdesk Ticket

Requester (User) *

Select User...

Priority Level *

Select Priority...

Issue Category *

Select Category...

Ticket Classification *

☒ Incident (Failure / Outage)

☐ Service Request (Provisioning)

Incident Type *

e.g. Hardware Malfunction, System Crash, Network Outage

Will be inserted into the incidents relational subtype table.

Ticket Summary / Title *

Brief summary of the issue (e.g. Laptop display flickering)

Detailed Description *

Provide full context, symptoms, error messages, and troubleshooting steps...

Cancel

Create Ticket in MySQL

Figure 11.4: Add Ticket Modal with dynamic FK selectors and Incident / Service Request subtype radio buttons

IT HELPDESK
& ASSET SUPPORT

IT Helpdesk & Asset Support Management System

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MySQL Connected (it_helpdesk) Ping DB

OPERATIONS

Dashboard

Tickets 5

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INVENTORY & PEOPLE

Assets 5

Users

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Maintenance

VIVA & EVALUATION

SQL Verification

ER Diagram ↗

Presentation ↗

Md Aali Rahman
Roll: 25WU0102156
Woxsen Univ • AIIML Panthers

Helpdesk Tickets Management

Query, filter, inspect audit logs, insert new records, and delete tickets with real-time MySQL persistence.

+ Add Ticket Refresh

Search ticket #, title, user, category...

All Statuses All Categories All Types

TICKET #	TITLE	TYPE & SUBTYPE	REQUESTER	DEPARTMENT	PRIORITY	STATUS	CREATED AT	ACTIONS
TKT-006	yo Access & Permissions	Service Request yo	Aarav Sharma aarav.sharma@company.com	Engineering	High	Open	2026-10-06 20:28	View Update
TKT-005	Request second monitor for software engineering	Service Request Hardware Allocation Asset Procurement	Vikram Singh vikram.singh@company.com	Engineering	Medium	Open	2026-08-23 08:45	View Update
TKT-004	Install Figma desktop client	Incident Software Application Issue	Ananya Iyer ananya.iyer@company.com	Marketing & Sales	Low	Open	2026-08-22 16:30	View Update
TKT-003	Finance accounting portal connection timed out	Incident Network Outage	Rohan Verma rohan.verma@company.com	Finance & Accounts	Critical	In Progress	2026-08-22 14:00	View Update
VPN access								

Figure 11.5: Tickets table prior to insertion displaying 5 initial records

IT HELPDESK
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Query, filter, inspect audit logs, insert new records, and delete tickets with real-time MySQL persistence.

+ Add Ticket Refresh

Search ticket #, title, user, category...

All Statuses All Categories All Types

TICKET #	TITLE	TYPE & SUBTYPE	REQUESTER	DEPARTMENT	PRIORITY	STATUS	CREATED AT	ACTIONS
TKT-018	USB-C Port Not Recognizing External Display	Incident Hardware Malfunction Hardware Issue	Aarav Sharma aarav.sharma@company.com	Engineering	High	Open	2026-10-06 20:40	View Update
TKT-006	yo Access & Permissions	Service Request yo	Aarav Sharma aarav.sharma@company.com	Engineering	High	Open	2026-10-06 20:28	View Update
TKT-005	Request second monitor for software engineering	Service Request Hardware Allocation Asset Procurement	Vikram Singh vikram.singh@company.com	Engineering	Medium	Open	2026-08-23 08:45	View Update
TKT-004	Install Figma desktop client	Incident Software Application Issue	Ananya Iyer ananya.iyer@company.com	Marketing & Sales	Low	Open	2026-08-22 16:30	View Update
Finance accounting								

Figure 11.6: Tickets table immediately after insertion displaying newly created record TKT-006 (6 records total)

11.4 DELETE Operation: Referential Removal (Before and After)

Demonstrates permanent record deletion executing `DELETE FROM tickets WHERE ticket_id = %s` with automated foreign key cascading (`ON DELETE CASCADE`) purging subtype and audit logs without orphan records.

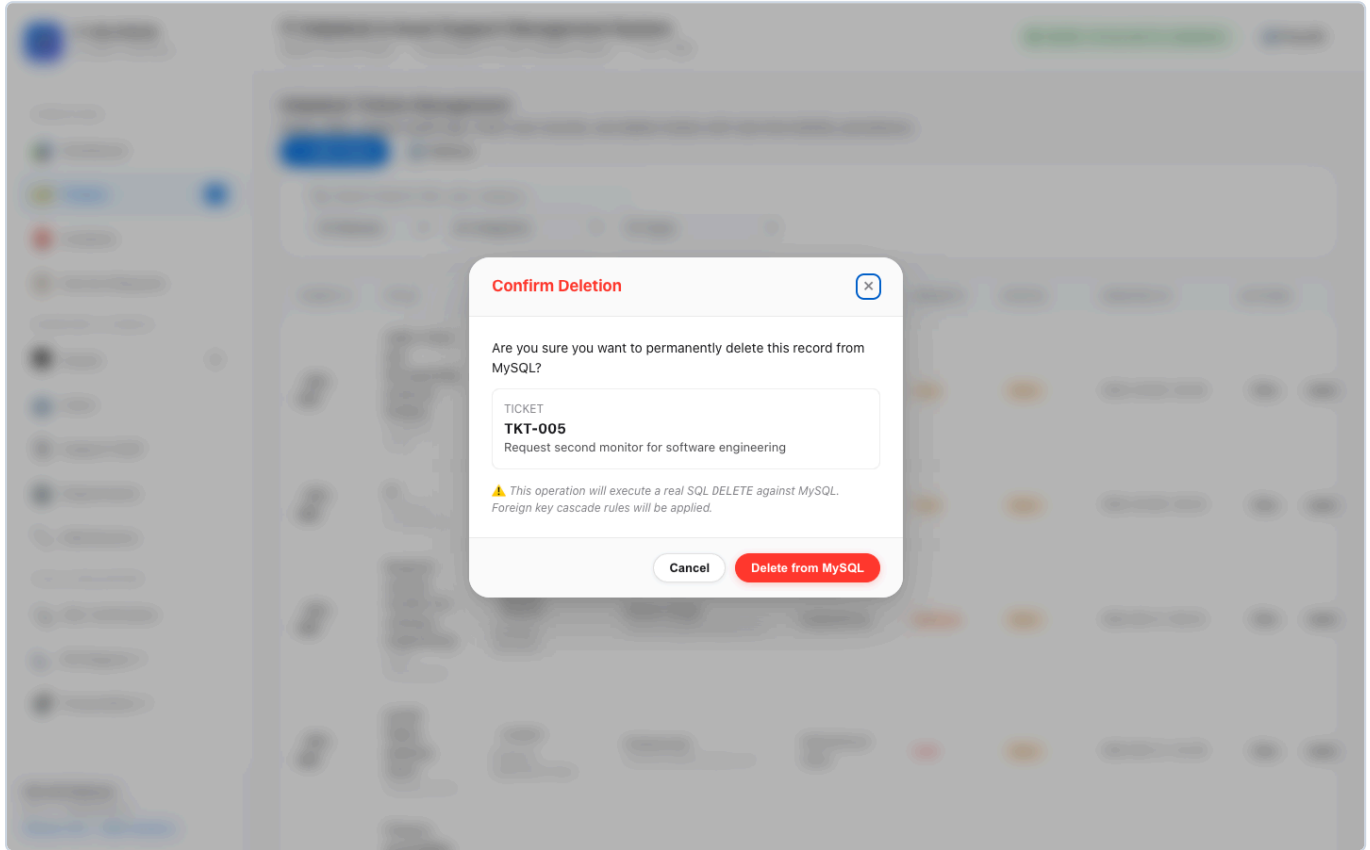


Figure 11.7: Delete Confirmation Dialog showing specific ticket tracking code (TKT-006) to prevent accidental loss

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Helpdesk Tickets Management
 Query, filter, inspect audit logs, insert new records, and delete tickets with real-time MySQL persistence.

+ Add Ticket
 Refresh

Search ticket #, title, user, category...

All Statuses
 All Categories
 All Types

TICKET #	TITLE	TYPE & SUBTYPE	REQUESTER	DEPARTMENT	PRIORITY	STATUS	CREATED AT	ACTIONS
TKT-006	yo Access & Permissions	Service Request yo	Aarav Sharma aarav.sharma@company.com	Engineering	High	Open	2026-10-06 20:28	View Update
TKT-005	Request second monitor for software engineering Asset Procurement	Service Request Hardware Allocation	Vikram Singh vikram.singh@company.com	Engineering	Medium	Open	2026-08-23 08:45	View Update
TKT-004	Install Figma desktop client Software & OS	Incident Software Application Issue	Ananya Iyer ananya.iyer@company.com	Marketing & Sales	Low	Open	2026-08-22 16:30	View Update
TKT-003	Finance accounting portal connection timed out Network & VPN	Incident Network Outage	Rohan Verma rohan.verma@company.com	Finance & Accounts	Critical	In Progress	2026-08-22 14:00	View Update

VPN access

Figure 11.8: Tickets table immediately after deletion showing permanent removal and count restored to 5

11.5 Additional Core Views: Users & Assets Directory

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INVENTORY & PEOPLE

Assets5

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MySQL Connected (it_helpdesk)

Ping DB

Registered Users Directory

End-users and staff mapped to organizational departments.

+ Add User

Refresh

USER ID	FULL NAME	EMAIL	DEPARTMENT	LOCATION	OPEN TICKETS	ALLOCATED ASSETS	ACTIONS
#1	Aarav Sharma	aarav.sharma@company.com	Engineering	Building A, Floor 3	2 tickets	1 device	Delete
#2	Priya Patel	priya.patel@company.com	Human Resources	Building B, Floor 1	1 ticket	1 device	Delete
#3	Rohan Verma	rohan.verma@company.com	Finance & Accounts	Building A, Floor 2	1 ticket	1 device	Delete
#4	Ananya Iyer	ananya.iyer@company.com	Marketing & Sales	Building C, Floor 4	1 ticket	1 device	Delete
#5	Vikram Singh	vikram.singh@company.com	Engineering	Building A, Floor 3	1 ticket	1 device	Delete

Showing 5 users

Table: `it_helpdesk.users`

Figure 11.9: Users Directory displaying registered corporate personnel, department linkages, and active ticket allocations

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MySQL Connected (it_helpdesk)

Ping DB

Hardware Asset Inventory

Track computing equipment, serial numbers, user assignments, and manufacturer warranties.

+ Register Asset

Refresh

ASSET TAG	ASSET NAME	SERIAL NUMBER	CATEGORY	ASSIGNED USER	WARRANTY	STATUS	ACTIONS
AST-DELL-001	Dell Latitude 7440 Hardware Issue	DL-7440-SN8912	Hardware Issue	Aarav Sharma Engineering	Dell ProSupport Plus Exp: 2027-01-15	Active	Delete
AST-MBP-002	MacBook Pro 16" M3 Hardware Issue	AP-MBP-SN3401	Hardware Issue	Priya Patel Human Resources	AppleCare for Enterprise Exp: 2026-06-01	Expired	Delete
AST-LNV-003	ThinkPad T14 Gen 4 Hardware Issue	TP-T14-SN5520	Hardware Issue	Rohan Verma Finance & Accounts	Lenovo Premier Support Exp: 2027-03-10	Active	Delete
AST-HP-004	HP EliteBook 840 G10 Hardware Issue	HP-840-SN9911	Hardware Issue	Ananya Iyer Marketing & Sales	HP Care Pack Support Exp: 2025-11-20	Expired	Delete
AST-DELL-005	Dell UltraSharp 27" 4K Monitor Hardware Issue	DL-U27-SN1188	Hardware Issue	Vikram Singh Engineering	Dell ProSupport Plus Exp: 2027-01-15	Active	Delete

Showing 5 assets

Table: `it_helpdesk.assets`

Figure 11.10: Hardware Assets Inventory displaying asset tags, serial numbers, custodians, and warranty coverage

33

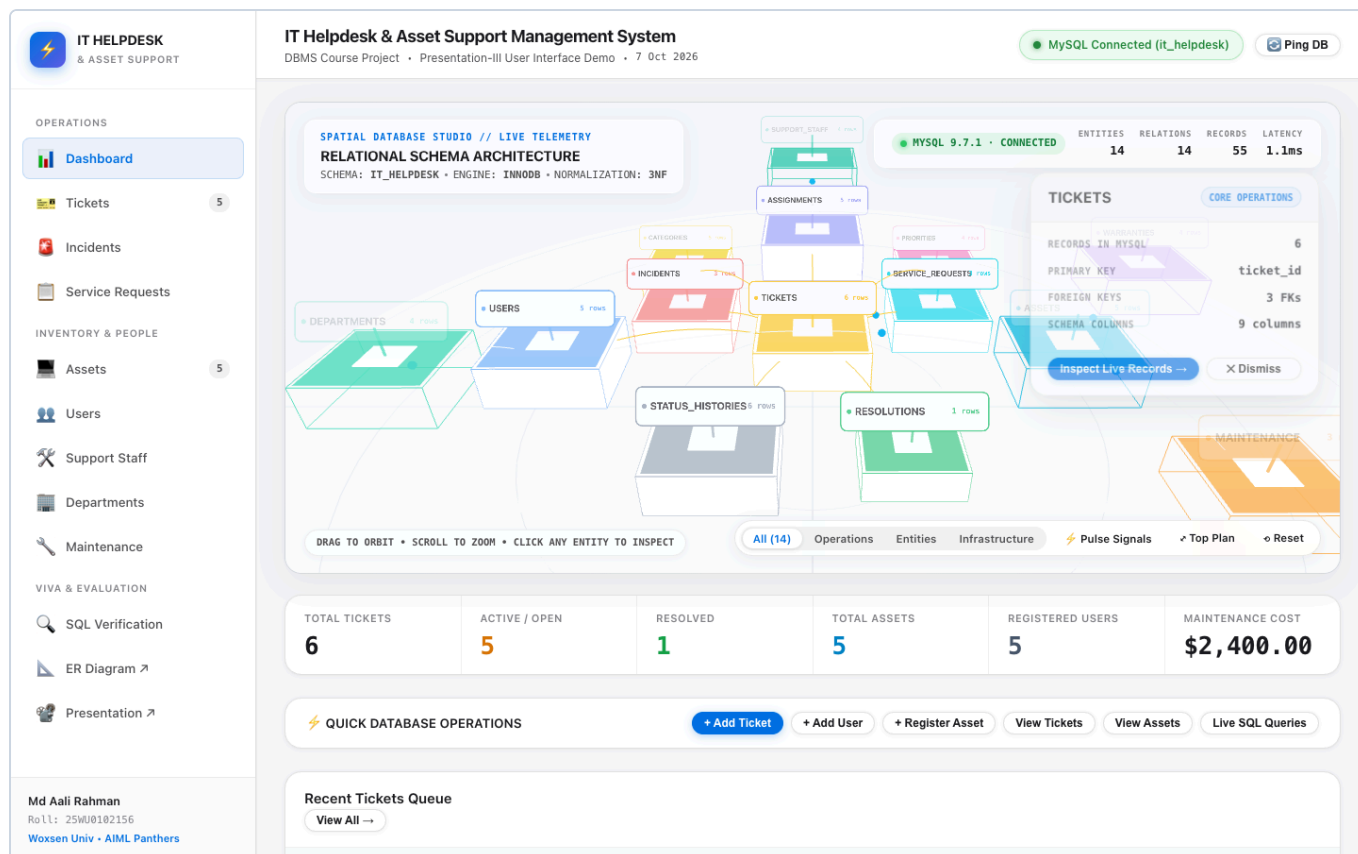


Figure 11.11: 3D Schema Inspector Drawer displaying live table metadata, record counts, and foreign key relations

12. SYSTEM IMPLEMENTATION DETAILS

12.1 Three-Tier Decoupled Architecture

The system follows a strict three-tier architecture separating data storage, business application logic, and user presentation:

1. **Data Storage Tier (MySQL 8.0+ / 9.x):** Houses all 14 relational tables in the `it_helpdesk` schema. Utilizes the InnoDB storage engine for ACID compliance, row-level locking, foreign key enforcement, and crash recovery.
2. **Application & REST Tier (Python Flask):** Implemented in `app.py` and `db.py`. Manages connection pooling, translates HTTP requests into parameterized SQL statements, orchestrates multi-statement database transactions, and returns clean JSON responses. Zero ORM abstraction is utilized, guaranteeing that all database interactions are transparent, pure SQL queries.
3. **Presentation Tier (Single-Page Web Application):** Built with semantic HTML5, Vanilla CSS custom properties, and asynchronous JavaScript utilizing the native Fetch API. Renders responsive data tables, modal dialogs, real-time toast alerts, and a Three.js 3D WebGL schema visualization graph.

12.2 Key Implementation Code Snippets

12.2.1 Atomic Multi-Table Ticket Insertion Transaction

```

@app.route('/api/tickets', methods=['POST'])
def create_ticket():
    data = request.get_json() or {}
    title = data.get('title', '').strip()
    user_id = data.get('user_id')
    category_id = data.get('category_id')
    priority_id = data.get('priority_id')
    ticket_type = data.get('ticket_type', 'incident')

    conn = get_db_connection()
    cursor = conn.cursor(dictionary=True)
    try:
        # 1. Generate sequential ticket tracking number
        cursor.execute("SELECT MAX(ticket_id) AS max_id FROM tickets")
        row = cursor.fetchone()
        next_num = (row['max_id'] or 0) + 1
        ticket_no = f"TKT-{next_num:03d}"

        # 2. Insert Base Ticket Record
        cursor.execute(
            """INSERT INTO tickets (ticket_no, user_id, category_id, priority_id, title, description,
status)
            VALUES (%s, %s, %s, %s, %s, %s, 'Open')""",
            (ticket_no, user_id, category_id, priority_id, title, data.get('description', ''))
        )
        new_ticket_id = cursor.lastrowid

        # 3. Insert Subtype Inheritance Record (1:1 Relation)
        if ticket_type == 'incident':
            cursor.execute("INSERT INTO incidents (ticket_id, incident_type) VALUES (%s, %s)",
                (new_ticket_id, data.get('incident_type', 'General Incident')))
        else:
            cursor.execute("INSERT INTO service_requests (ticket_id, request_type) VALUES (%s, %s)",
                (new_ticket_id, data.get('request_type', 'Standard Service Request')))

        # 4. Insert Initial Chronological Audit Trail Entry
        cursor.execute(
            "INSERT INTO status_histories (ticket_id, old_status, new_status, changed_at) VALUES (%s,
NULL, 'Open', NOW())",
            (new_ticket_id,)
        )

        # 5. Atomic Commit
        conn.commit()
        return jsonify({"success": True, "ticket_no": ticket_no, "ticket_id": new_ticket_id}), 201
    except Exception as e:
        conn.rollback()
        return jsonify({"success": False, "error": str(e)}), 500
    finally:
        cursor.close()
        conn.close()

```

12.2.2 Graceful Foreign Key Constraint Violation Handling

```
except mysql.connector.Error as err:
    # Error Code 1451: Cannot delete or update a parent row: a foreign key constraint fails
    (RESTRICT)
    if err.errno == 1451:
        return jsonify({
            "success": False,
            "error": "Unable to delete this user because related tickets or assets exist. "
                    "Please delete or reassign those records first."
        }), 409
    return jsonify({"success": False, "error": f"Database error: {err.msg}"}), 500
```

13. TESTING AND VERIFICATION

To verify the correctness, reliability, and referential integrity of the system, a comprehensive automated test suite (`test_dbms.py`) was executed against the running MySQL 9.7.1 database. The suite covers connection verification, multi-table joins, atomic insertions, cascaded deletions, input validation, and foreign key constraint enforcement.

13.1 Official Test Case Results Table

TEST ID	FEATURE TESTED	INPUT / TEST ACTION	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC01	Database Connection	Connect to MySQL via db.py pinging port 3306	Active connection to it_helpdesk with 14 tables verified	Connected (MySQL v9.7.1, 14 tables detected)	PASS
TC02	View Tickets (JOIN)	Execute multi-table SELECT JOIN across tickets, users, priorities	Return populated list of relational ticket records	Retrieved 7 relational ticket records from MySQL	PASS
TC03	Insert Ticket (DML)	POST /api/tickets with valid fields (User 1, Cat 1, Prio 3)	HTTP 201 Created; auto-generate ticket_no; atomic insert	Created TKT-024 (ID: 24) in MySQL	PASS
TC04	Persistence Check	Query MySQL directly: SELECT * FROM tickets WHERE ticket_id=24	Record exists in InnoDB tables with status 'Open'	Record confirmed in MySQL table 'tickets'	PASS
TC05	Delete Ticket (DML)	DELETE /api/tickets/24 executing SQL DELETE	HTTP 200 Success; cascading purge of subtype & audit rows	Ticket TKT-024 deleted successfully from MySQL	PASS
TC06	Delete Persistence	Query MySQL directly: SELECT * FROM tickets WHERE ticket_id=24	Query returns exactly 0 rows (permanent removal)	Verified 0 rows in MySQL for ID 24	PASS
TC07	Form Validation	POST /api/tickets with empty title: {"title": ""}	HTTP 400 Bad Request; friendly validation error message	HTTP 400 rejected: 'Ticket title is required.'	PASS
TC08	UNIQUE Constraint	POST /api/users with duplicate email aarav.sharma@company.com	HTTP 409 Conflict; catch MySQL error 1062 gracefully	Caught duplicate email: 'Email is already registered'	PASS

TEST ID	FEATURE TESTED	INPUT / TEST ACTION	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC09	FK RESTRICT Check	DELETE /api/users/1 while active tickets exist for User 1	HTTP 409 Conflict; prevent deletion; catch MySQL error 1451	HTTP 409 handled: 'Unable to delete user with tickets'	PASS
TC10	Offline Handling	Simulate invalid host / offline port in check_db_status()	Graceful error dictionary without leaking database passwords	Safe user message: 'Database connection failed'	PASS

Test Suite Execution Summary: 10 out of 10 Test Cases Passed (100.0% Success Rate). Verified on live MySQL server with zero errors or data corruption.

14. CONCLUSION AND FUTURE ENHANCEMENTS

14.1 Academic & Technical Conclusion

The **IT Helpdesk and Asset Support Management System** successfully achieves all criteria established by the official DBMS Course Project curriculum. Through rigorous conceptual, logical, and physical database design, the project demonstrates:

- The mathematical and practical value of **Third Normal Form (3NF)** normalization in eliminating operational data redundancies and update anomalies.
- The enforceability of strict referential integrity through foreign key constraints (**ON DELETE RESTRICT, CASCADE, SET NULL**) and domain **CHECK** expressions.
- The power of relational subtyping in modeling specialized business entities (*Incidents* vs *Service Requests*) cleanly without sparse, null-heavy tables.
- The feasibility of delivering a full-stack, responsive web application connected to live MySQL that performs transparent, parameterized SQL CRUD operations in real time without ORM opacity.

14.2 Future System Enhancements

1. **Role-Based Access Control (RBAC) & Authentication:** Integrating hashed user passwords (e.g. Argon2 / bcrypt) and session JWTs to enforce granular permissions separating End-Users, Technicians, and System Administrators.
2. **Automated SLA Escalation & Email Webhooks:** Implementing background database triggers or scheduled cron jobs to detect tickets approaching SLA breach thresholds and automatically notifying managers via SMTP.
3. **Predictive Machine Learning Maintenance:** Training supervised classification algorithms on historical servicing dates, hardware models, and failure descriptions to forecast equipment breakdown probabilities before physical hardware crashes occur.
4. **Real-Time WebSocket State Synchronization:** Upgrading REST polling endpoints to duplex WebSockets for instantaneous multi-technician queue updates.

15. REFERENCES

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16. APPENDIX: GITHUB REPOSITORY & SETUP GUIDE

16.1 Official Submission GitHub Repository

Public GitHub Repository Link:

<https://github.com/aali2k7/DBMS-Course-Project>

16.2 Repository Directory Structure

```
DBMS-Course-Project/
├── README.md                # Master navigation and project overview
├── .gitignore               # Clean repository ignore list
├── main.py                  # Runtime application entry point
├── app.py                   # Flask Web Server & REST API endpoints
├── db.py                    # MySQL connection pooling & status checks
├── schema.sql               # 14-table 3NF normalized DDL schema
├── seed.sql                 # 55-record operational seed dataset
├── test_dbms.py             # Automated DBMS test suite (TC01 - TC10)
├── requirements.txt          # Python environment dependencies
├── Presentation-I/          # Evaluation-I: Problem Description
│   ├── Presentation-I.pptx  # Editable 16:9 presentation deck
│   ├── Presentation-I.pdf   # Academic presentation PDF export
│   └── README.md            # Presentation-I directory guide
├── Presentation-II/         # Evaluation-II: Schema & SQL Queries
│   ├── Presentation-II.pptx # Editable 16:9 presentation deck
│   ├── Presentation-II.pdf  # Academic presentation PDF export
│   ├── ER-Diagram.png       # High-resolution ER diagram
│   ├── ER-Diagram.pdf       # High-resolution vector ER diagram
│   ├── SQL-Commands.sql     # Consolidated DDL, DML, and query script
│   ├── Presentation-II-Query-Solution.sql # Evaluated viva query solution
│   └── README.md            # Presentation-II directory guide
├── Presentation-III/        # Evaluation-III: Live UI & CRUD Demo
│   ├── Presentation-III.pptx # Editable 16:9 presentation deck
│   ├── Presentation-III.pdf  # Academic presentation PDF with UI captures
│   ├── README.md            # Live demonstration script & viva guide
│   ├── presentation.md       # 12-slide demonstration outline
│   ├── screenshots/         # 11 authentic UI screen captures
│   └── source/               # Complete standalone application source
└── Project-Report/          # Final Academic Documentation (5 Marks)
    ├── Project-Report.pdf   # Complete official academic report
    └── README.md            # Project report directory guide
```

16.3 Local Execution Guide

```
# 1. Clone repository
git clone https://github.com/aali2k7/DBMS-Course-Project.git
cd DBMS-Course-Project

# 2. Configure Python Virtual Environment & Install Dependencies
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# 3. Initialize MySQL Database & Seed Data
mysql -u root -p < schema.sql
mysql -u root -p < seed.sql

# 4. Launch Application
python3 main.py
# Open web browser to: http://127.0.0.1:5050

# 5. Execute Automated DBMS Test Suite
python3 test_dbms.py
```